Bayesian Networks

This section explores tools used for modelling, problem representation, and inferencing using Bayesian Networks along with a simple example implementation.

Contents

A Bayesian Network is a probabilistic graphical model that represents random variables and their conditional dependencies using a directed acyclic graph (DAG).

Bayesian Networks are used to model uncertainty and perform probabilistic reasoning.

They are widely used in:

- Medical diagnosis
- Weather prediction
- Risk analysis

In a Bayesian Network:

- Nodes represent variables
- Edges represent dependencies
- Probabilities define relationships between variables

Example:

Rain → Wet Grass
Rain → Traffic

Components of Bayesian Networks

1. Nodes

Nodes represent random variables.

Examples:

- Rain
- Traffic
- Disease
- Fever

2. Edges

Edges represent probabilistic dependencies between variables.

Example:

Rain → Traffic

Meaning traffic depends on rain conditions.

3. Conditional Probability Table (CPT)

Each node contains probabilities based on parent nodes.

Example:

Rain	Traffic Heavy
True	0.8
False	0.2

Tools for Bayesian Networks

1. GeNIe

GeNIe is a graphical tool for building Bayesian Networks.

Features:

- Drag-and-drop network creation
- Probabilistic inference
- Decision analysis

- Visualization support

Applications:

- Medical diagnosis
- Risk management

2. Netica

Netica is a software tool used for probabilistic reasoning and Bayesian Network modelling.

Features:

- Bayesian inference
- Graph visualization
- Decision support systems

Applications:

- AI systems
- Fault analysis

3. pgmpy

pgmpy is a Python library for probabilistic graphical models.

Features:

- Bayesian Network modelling
- Probabilistic inference
- Learning algorithms

Applications:

- Machine learning
- AI research

4. BayesiaLab

BayesiaLab is a commercial tool for advanced Bayesian analysis.

Features:

- Data analytics

- Predictive modelling
- Knowledge discovery

Applications:

- Healthcare
- Finance
- Industrial AI

Inferencing in Bayesian Networks

Inference is the process of calculating probabilities of unknown variables using known evidence.

Types of inference:

- Exact inference
- Approximate inference

Example:

P(Rain | Wet Grass)

This calculates the probability of rain given that the grass is wet.

Common inference algorithms:

- Variable Elimination
- Belief Propagation
- Sampling Methods

Example Problem

Medical Diagnosis System

Consider a simple Bayesian Network:

Disease → Fever
Disease → Cough

Variables

- Disease
- Fever
- Cough

Conditional Probabilities

Disease	Fever
True	0.9
False	0.2

Disease	Cough
True	0.8
False	0.3

g++ -std=c++11 bayesian.cpp -o bayesian
./bayesian

```
(base) rangoju@Rangojus-MacBook-Pro Q4 % nano bayesian.cpp
(base) rangoju@Rangojus-MacBook-Pro Q4 % g++ -std=c++11 bayesian.cpp -o bayesian
(base) rangoju@Rangojus-MacBook-Pro Q4 % ./bayesian
Probability of Disease given Fever and Cough: 0.108108
(base) rangoju@Rangojus-MacBook-Pro Q4 %
```

The example implementation demonstrates how Bayesian inference can be used to calculate probabilities based on observed evidence.